

SOWGOTO RAHA SUNNY

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Academic Profile

M.S. graduate in Cybersecurity with a GPA of 3.72/4.00 and research experience in machine and deep learning, explainable and trustworthy AI, anomaly detection, network security, and privacy-preserving federated learning. Research interests include AI-driven cybersecurity, secure and reliable intelligent systems, large language models (LLMs), and distributed AI. Seeking Ph.D. opportunities to develop robust, interpretable, and privacy-aware learning methods for secure networked systems, edge intelligence, and autonomous systems.

Education

Master of Science in Cybersecurity <i>University of North Texas, Denton, TX</i>	2026 GPA: 3.72/4.00
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Master of Science in Information Technology <i>Jahangirnagar University, Dhaka, Bangladesh</i>	2019 GPA: 3.01/4.00
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Bachelor of Science in Information Technology <i>Jahangirnagar University, Dhaka, Bangladesh</i>	2017 GPA: 3.45/4.00
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Research Interests

Trustworthy and secure artificial intelligence, machine learning and deep learning, explainable AI, anomaly detection, privacy-preserving and distributed learning, reliable edge intelligence, autonomous multi-agent systems, and the security and resilience of intelligent networked systems.

Research Experience

Explainable Deep Learning for Network Intrusion Detection <i>Independent Research</i>	2025 – Present Texas, USA
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- Developed NML-XAI combining Transformer-based feature processing, CNN-BiLSTM-attention-autoencoder models, and ensemble stacking for explainable network intrusion detection.
- Applied SMOTE, SHAP, and LIME to UNSW-NB15 and achieved 93.13% accuracy, 93.45% precision, and 93.19% F1-score.
- Investigated interpretable machine-learning techniques for identifying anomalous network behavior and improving confidence in automated cybersecurity decisions.

Hybrid KAN and Explainable AI for DDoS Detection <i>Collaborative Research Project</i>	2025 – 2026 Texas, USA
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- Designed a hybrid Kolmogorov–Arnold Network and explainable AI pipeline for multi-class DDoS detection in software-defined networks.
- Evaluated predictive performance and interpretable feature behavior to improve reliable security decisions in networked systems.
- Contributed to experimental design, model evaluation, technical analysis, manuscript preparation, and research documentation.

Publications and Manuscripts

1. **S. R. Sunny**, Z. A. Mamun, “HKAN-XAI: A Hybrid Kolmogorov–Arnold Network with Explainable AI for Multi-Class DDoS Attack Detection in SDN,” accepted for oral presentation at the *2nd IEEE International Conference on Cybersecurity and AI-Based Systems (Cyber-AI 2026)*, Bucharest, Romania, 2026.
2. **S. R. Sunny**, S. S. Adhimulam, and P. R. Ovi, “The Future of Database Resilience: A Systematic Literature Review on Mitigating AI-Driven Ransomware in Transactional Systems,” manuscript under review.
3. M. Islam, T. Mohiuddin, M. A. Rahman, Md. Sharfuddin, Md Sazidul Islam, **S. R. Sunny**, and E. Lokesh, “A Hybrid Deep Learning Framework for Privacy-Preserving Sepsis Prediction in Distributed ICU Environments Using Federated Learning Simulation,” *Discover Social Science and Health*, published 28 September 2026. doi:10.1007/s44155-026-00489-1.

Teaching Experience

Graduate Teaching Assistant – Applied Machine Learning <i>University of North Texas</i>	Jan 2025 – May 2025 Denton, TX
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- Supported graduate-level instruction in applied machine learning, programming, and computational problem solving.
- Provided office-hour and laboratory support for Python, software configuration, debugging, code review, and technical assignments.
- Guided students on regression, clustering, decision trees, random forests, SVMs, neural networks, and model evaluation.
- Created technical documentation and explained solutions to students with varied programming and technical backgrounds.

Professional Engineering Experience

Senior System Engineer

Dotlines Bangladesh, Dhaka, Bangladesh

Jul 2022 – Jul 2024

- Engineered and administered Linux and Windows production environments supporting servers, databases, APIs, web services, telecommunications, and transactional systems.
- Maintained 99.9% service availability across platforms processing more than 50,000 daily transactions through monitoring, troubleshooting, automation, and incident response.
- Developed Python, Bash, SQL, PL/SQL, and PowerShell automation for diagnostics, job monitoring, data analysis, and service-health validation.
- Reduced mean time to resolution by 30% across 35+ monthly incidents using Linux log analysis, database investigation, API validation, and structured root-cause analysis.
- Collaborated with development, DevOps, cybersecurity, database, and infrastructure teams on CI/CD deployments, rollback validation, testing, and production troubleshooting.
- Performed vulnerability assessment, security remediation, access reviews, backup verification, and system-hardening activities.

System Engineer

Dotlines Bangladesh, Dhaka, Bangladesh

May 2019 – Jun 2022

- Supported Linux/Windows servers, databases, enterprise applications, networking, payment gateways, VAS, bulk SMS/OTP, and telecommunications services.
- Resolved 60+ weekly incidents involving application errors, SQL/database issues, Linux services, network connectivity, user access, file transfers, and UAT validation while meeting SLA targets.
- Performed server maintenance, patch validation, backup verification, access reviews, security remediation, and service-health monitoring.
- Created SOPs, troubleshooting guides, runbooks, RCA records, and technical documentation to support engineering and operations teams.

Selected Research and Technical Projects

Ransomware Attack Simulation and Mitigation — Developed a Python-based monitoring and mitigation environment using Watchdog, psutil, and SQLite logging to study ransomware behavior and defensive controls.

AI-Enhanced Zero Trust Cybersecurity — Designed a cloud-oriented Zero Trust architecture incorporating continuous authentication, anomaly detection, least-privilege access, and automated policy enforcement.

Web Vulnerability Scanner Analysis — Evaluated OpenVAS, Nikto, and Wapiti against OWASP-relevant targets, comparing vulnerability-detection capability and false-positive behavior.

Technical Skills

Programming & Scripting: Python, Java, SQL, PL/SQL, Bash/Shell, PowerShell, C/C++

Machine Learning: Transformers, CNN, RNN, BiLSTM, autoencoders, ensemble learning, SVM, SMOTE, model evaluation
Explainable / Trustworthy AI: SHAP, LIME, explainability, anomaly detection, uncertainty-aware and reliability-focused analysis

Systems: Linux (Ubuntu, RHEL), UNIX, Windows Server, Windows 10/11, macOS

Cloud & DevOps: AWS, Microsoft Azure, Docker, Kubernetes, VMware, Hyper-V, Git/GitHub, Jenkins, CI/CD, Terraform, Ansible

Databases: MySQL, MariaDB, SQL Server, Oracle, PostgreSQL, ETL pipelines, batch processing

Networking & Security: TCP/IP, DNS, DHCP, VPN, Wireshark, vulnerability assessment, network security, access control, incident response

Teaching: Python instruction, laboratory support, code review, debugging, student mentoring, technical documentation, instructional technology

Certification

AWS Certified Solutions Architect – Associate | Credential Link

June 2026